Decision Analysis Theory

Decision Making Under Risk and Uncertainty 2

Announcements

- Homework 2 – Assigned 02/12
  - Due 02/19
- Project – Problem Statement – Assigned 02/02
  - Due 02/17
- Extra Credit – Decision Analysis example – Due Today

Outline

- Decision Without Experimentation
  - Alternatives
  - Prior Probabilities
  - Bayes Decision Rule
- Decision With Experimentation
  - Seismic Survey
  - Posterior Probabilities
  - Value of Experimentation
- Decision Tree Analysis
- Utility Theory

Decision Without Experimentation

- The Oildrill Company owns land that may contain oil.
- A consulting geologist has reported to management that she believes there is one chance in four of oil.
- Because of this prospect, another oil company has offered to purchase the land for $90,000.
- However, Oildrill is considering holding the land in order to drill for oil itself.
- The cost of drilling is $100,000. If oil is found, the resulting expected revenue will be $800,000, so the company's expected profit (after deducting the cost of drilling) will be $700,000. A loss of $100,000 (the drilling cost) will be incurred if the land is dry (no oil).

Decision Without Experimentation

- Prospective profits for the Oildrill Company.

Alternatives	**Futures/ Probability**	
	Oil 0.25	Dry 0.75
Drill for oil	$700,000	-$100,000
Sell the land	$90,000	$90,000

Decision Without Experimentation

- Bayes Decision Rule.
  - Using the best available estimates of the probabilities of the respective states of nature (currently the prior probabilities), calculate the expected value of the payoff for each of the possible alternatives and choose the one with the maximum expected payoff.

  - *E*[Payoff (drill)] = 0.25($700,000) + 0.75(-$100,000) = $100,000
  - *E*[Payoff (sell)] = 0.25($90,000) + 0.75($90,000) = $90,000
  - Select to drill for oil.

Decision With Experimentation

- Posterior Probabilities.
  - Additional testing (experimentation) can be done to improve the preliminary estimates of the probabilities of the respective states of nature provided by the prior probabilities.

- Identify possible experimentation.
- Derive posterior probabilities.
- Decide whether is worthwhile to conduct experimentation.

Decision With Experimentation

- Before deciding whether to drill or sell, another option is to conduct a detailed seismic survey of the land to obtain a better estimate of the probability of finding oil. The cost of the seismic survey is $30,000.
- The seismic survey obtains seismic soundings to indicate whether the geological structure is favorable to the presence of oil. The possible findings of the survey are divided into the following categories:
  - Unfavorable: unfavorable seismic soundings; oil is fairly unlikely.
  - Favorable: favorable seismic soundings; oil is fairly likely.
- Based on past experience, if there is oil, the probability of unfavorable seismic soundings is 0.4; if there is no oil, the probability of unfavorable seismic soundings is 0.8.

Decision With Experimentation

- If there is oil:
  - P(Unfavorable | State = Oil) = 0.4
  - P(Favorable | State = Oil) = 1 – 0.4 = 0.6
- If there is no oil:
  - P(Unfavorable | State = Dry) = 0.8
  - P(Favorable | State = Dry) = 1 – 0.8 = 0.2

Posterior Probabilities

- Let n = number of possible states of natures.
- P(State = state i) = prior probability that true state of nature is state i, for i = 1, 2,… , n.
- Let Finding = finding from experimentation (a random variable).
- Finding j = one possible value of finding.
- P(State = state i | Finding = finding j) = posterior probability that true state of nature is state i, given that Finding = finding j, for i = 1, 2,…, n.

Bayes Theorem

- Given two events A and B, with $P(B) \neq 0$, then
  $$P(A|B) = \frac{[P(B|A)P(A)]}{P(B)}$$

- If A_k is a partition of A, then $P(B) = \sum P(B|A_k)P(A_k)$
  or
- $P(B) = P(B|A)P(A) + P(B|\neg A)P(\neg A)$, where
  $\neg A$ = complement of A

Posterior Probabilities

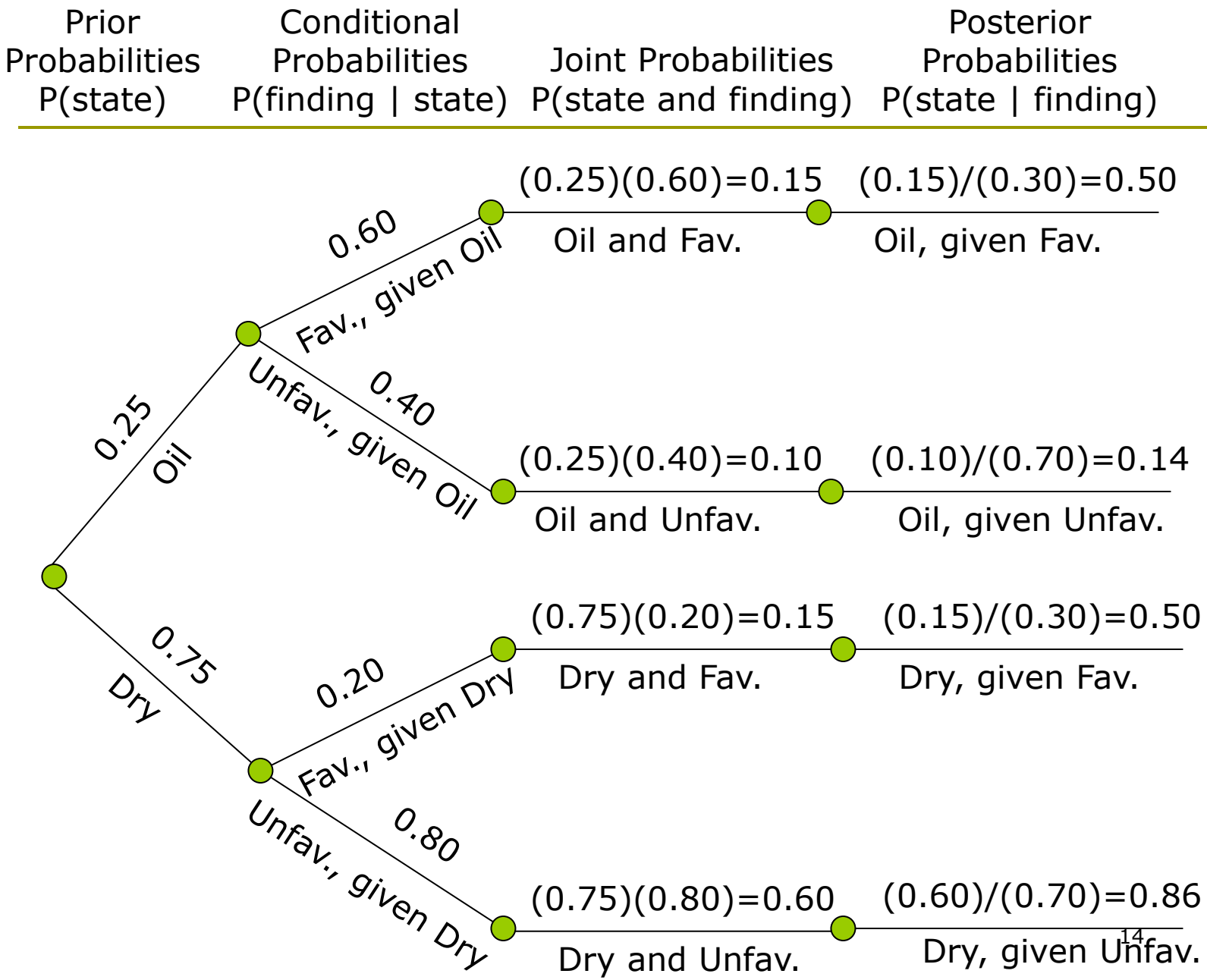
- Let *A* be the event: State = state *i*
- Let *B* be the event: Finding = finding *j*
- Then,
  $$P(State = state\ i \mid Finding = finding\ j) = \frac{[P(Finding = finding\ j \mid State = state\ i)P(State = state\ i)]}{P(Finding = finding\ j)}$$
- For State = state *k*, then *P*(Finding = finding *j*) = ∑*P*(*P*(Finding = finding *j* | State = state *k*)*P*(State = state *k*)
- Then, *P*(State = state *i* | Finding = finding *j*) = [*P*(Finding = finding *j* | State = state *i*)*P*(State = state *i*)] / ∑*P*(*P*(Finding = finding *j* | State = state *k*)*P*(State = state *k*)

Decision With Experimentation

- If the finding of the seismic survey is unfavorable:
  - P(State = Oil | Finding = Unfavorable) = [(0.4)(0.25)] / [(0.4)(0.25) + (0.8)(0.75)] = 1/7 = 0.14
  - P(State = Dry | Finding = Unfavorable) = 1 – 1/7 = 6/7 = 0.86
- If the finding of the seismic survey is favorable:
  - P(State = Oil | Finding = Favorable) = [(0.6)(0.25)] / [(0.6)(0.25) + (0.2)(0.75)] = 1/2 = 0.5
  - P(State = Dry | Finding = Favorable) = 1 – 1/2 = 1/2 = 0.5

Probability Tree Diagram

P(Favorable) = 0.15 + 0.15 = 0.30
P(Unfavorable) = 0.10 + 0.60 = 0.70



Decision With Experimentation

- Bayes Decision Rule.
- Expected payoffs if finding is unfavorable:
  - *E*[Payoff (drill) | Finding = Unfavorable] = (1/7)($700,000) + (6/7)(-$100,000) – $30,000 = -$15,700
  - *E*[Payoff (sell) | Finding = Unfavorable] = (1/7)($90,000) + (6/7)($90,000) – $30,000 = $60,000
  - Select to sell the land.
- Expected payoffs if finding is favorable:
  - *E*[Payoff (drill) | Finding = Favorable] = (1/2)($700,000) + (1/2)(-$100,000) – $30,000 = $270,000
  - *E*[Payoff (sell) | Finding = Favorable] = (1/2)($90,000) + (1/2)($90,000) – $30,000 = $60,000
  - Select to drill for oil.

Decision With Experimentation

- There is no information whether to conduct the experimentation.

Finding from experimentation	Optimal alternative	Expected payoff excluding cost of experimentation	Expected payoff including cost of experimentation
Unfavorable	Sell the land	$90,000	$60,000
Favorable	Drill for oil	$300,000	$270,000

- It may be better to forego the experimentation expense and just use the optimal solution without experimentation
  - Drill for oil with an expected payoff of $100,000.

The Value of Experimentation

- Before performing any experiment, it is necessary to determine its potential value.
  - Allows us to test assumptions, gather evidence, and gain deeper understanding by observing real-world effects of our ideas and actions
- Enables:
  - Tests assumptions
  - Builds understanding
  - Advances knowledge
  - Drives innovation
  - Fosters informed decisions
  - Connects theory to reality

The Value of Experimentation

- Two step process:
  - Expected value of perfect information (EVPI)
  - Expected value of experimentation (EVE)
    - Also known as Expected Value of Sample Information (EVSI)

The Value of Experimentation

- Expected value of perfect information.
  - Assume that the experiment will remove all uncertainty about what the true state of nature is.
  - Choose the action with the maximum payoff for the state.
  - Since the true state is not known in advance, calculate the expected payoff with perfect information (ignoring the cost of experimentation).
  - Provides an upper bound on the potential value of the experiment.
  - If the upper bound is less than the cost of experiment, the experiment should not be made.
  - If the upper bound exceeds the cost of experiment, then the experiment may be worthwhile performing.

The Value of Experimentation

- Expected value of perfect information.

Alternative	State of Nature	
	Oil	Dry
Drill for oil	$700,000	-$100,000
Sell the land	$90,000	$90,000
Maximum payoff	$700,000	$90,000
Prior probability	0.25	0.75

  - Expected payoff with perfect information:

    (0.25)($700,000) + (0.75)($90,000) = $242,500

  - If the company could learn before choosing its action whether the land contains oil, the expected payoff before acquiring the information would be $242,500 (excluding the cost of experimentation).

The Value of Experimentation

- Expected value of perfect information.
  - Calculation of this measure (EVPI)
  - EVPI = expected payoff with perfect information – expected payoff without experimentation.
  - Since experimentation cannot usually provide perfect information, EVPI provides an upper bound on the expected value of experimentation.

    EVPI = $242,500 - $100,000 = $142,500

  - Since EVPI exceeds cost of experimentation ($30,000) it may be worthwhile to proceed with the experiment.

The Value of Experimentation

- Expected value of experimentation.
  - Also called expected value of sample information.
  - Calculate the actual improvement in the expected payoff (ignoring the cost of the experiment) that would result from performing the experiment.
  - Comparing this improvement with the cost of experimentation indicates whether the experiment should be performed or not.

The Value of Experimentation

- Expected value of experimentation.
  - Determine all posterior probabilities.
  - Determine resulting optimal policy with experimentation.
  - Determine the corresponding expected payoff for each possible finding from the experiment.

  - Expected payoff with experimentation =

$$\sum P(\text{Finding} = \text{finding } j)E[\text{payoff} \mid \text{Finding} = \text{finding } j]$$

The Value of Experimentation

- Expected value of experimentation.
  - P(Unfavorable) = 0.70 and P(Favorable) = 0.30
  - *E*[Payoff | Finding = Unfavorable] = $90,000
  - *E*[Payoff | Finding = Favorable] = $300,000
  - Expected payoff with experimentation =

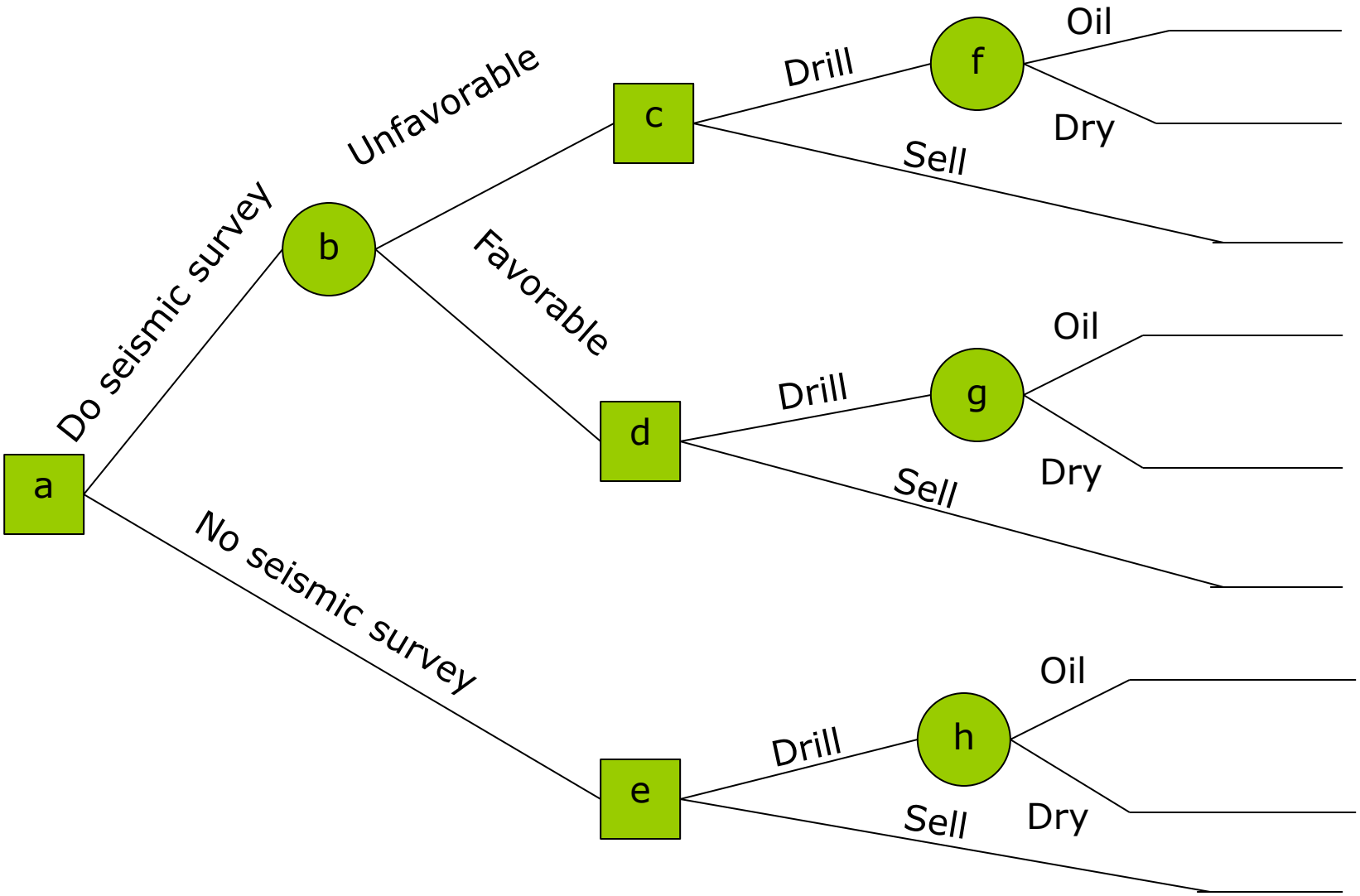
    (0.70)($90,000) + (0.30)($300,000) = $153,000

- Expected Value of Experimentation (EVE) = expected payoff with experimentation - expected payoff without experimentation
  - EVE = $153,000 - $100,000 = $53,000
  - Since EVE exceeds cost of experimentation ($30,000) the company should proceed with the experimentation.

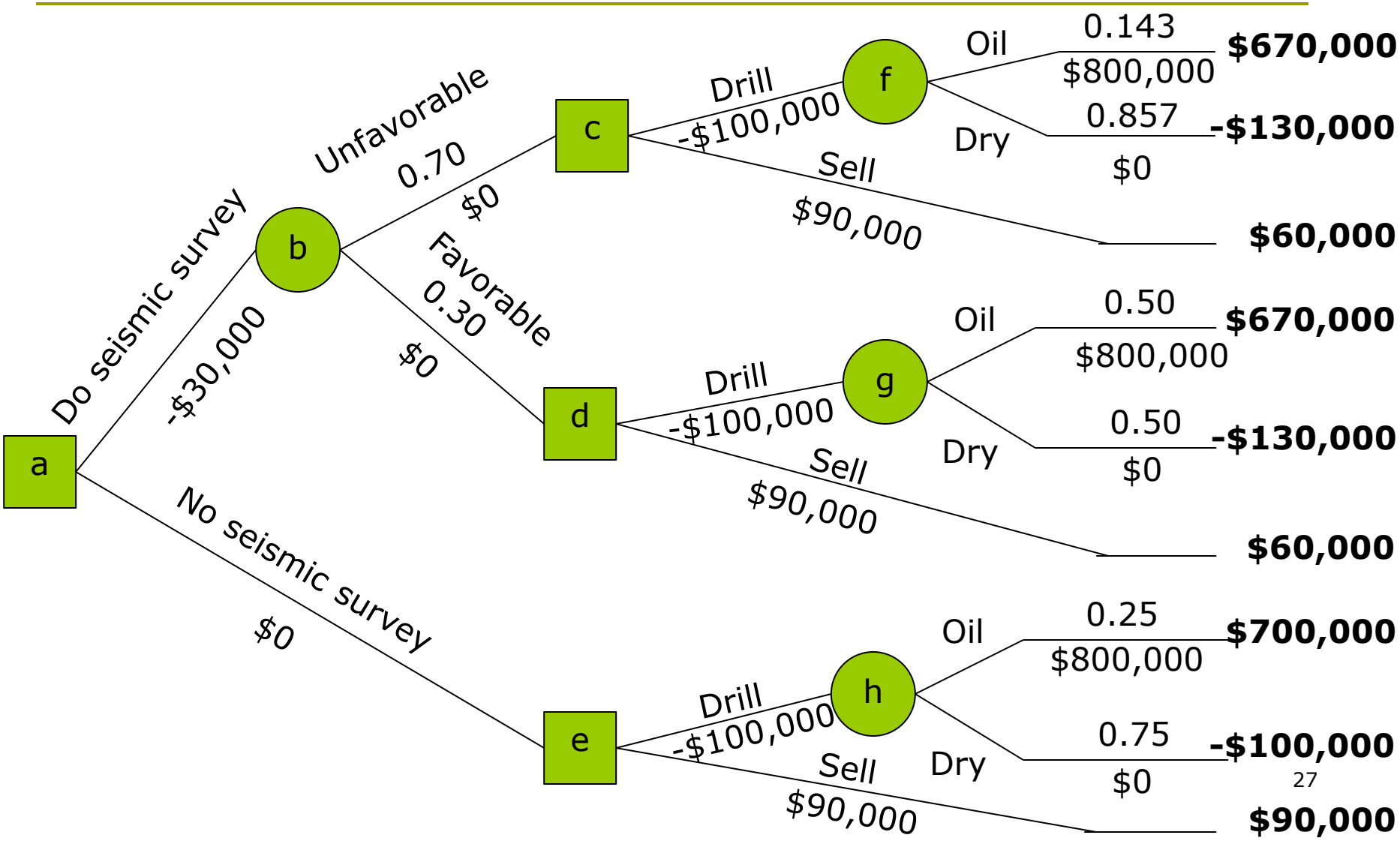
Decision Trees

- Decision trees provide a useful way of visually displaying the problem and then organizing the computational work.
  - Especially helpful when a sequence of decisions must be made.
- Junction points (nodes).
  - Decision nodes.
    - Indicates that a decision needs to be made at that point in the process.
  - Event nodes (state-of-nature nodes).
    - Indicates that a random event occurs at that point.
- Branches.
  - Lines between the junction points.

Decision Tree Analysis



Decision Tree Analysis



Decision Tree Analysis

- Procedure:
  - 1. Start at the right side of the decision tree and move left one column at a time. For each column, perform either step 2 or step 3 depending upon whether the nodes in that column are event nodes or decision nodes.
  - 2. For each event node, calculate its *expected payoff* by multiplying the expected payoff of each branch (shown in bold to the right of the branch) by the probability of that branch and then summing these products. Record this expected payoff for each decision node in bold next to the node, and designate this quantity as also being the expected payoff for the branch leading to this node.

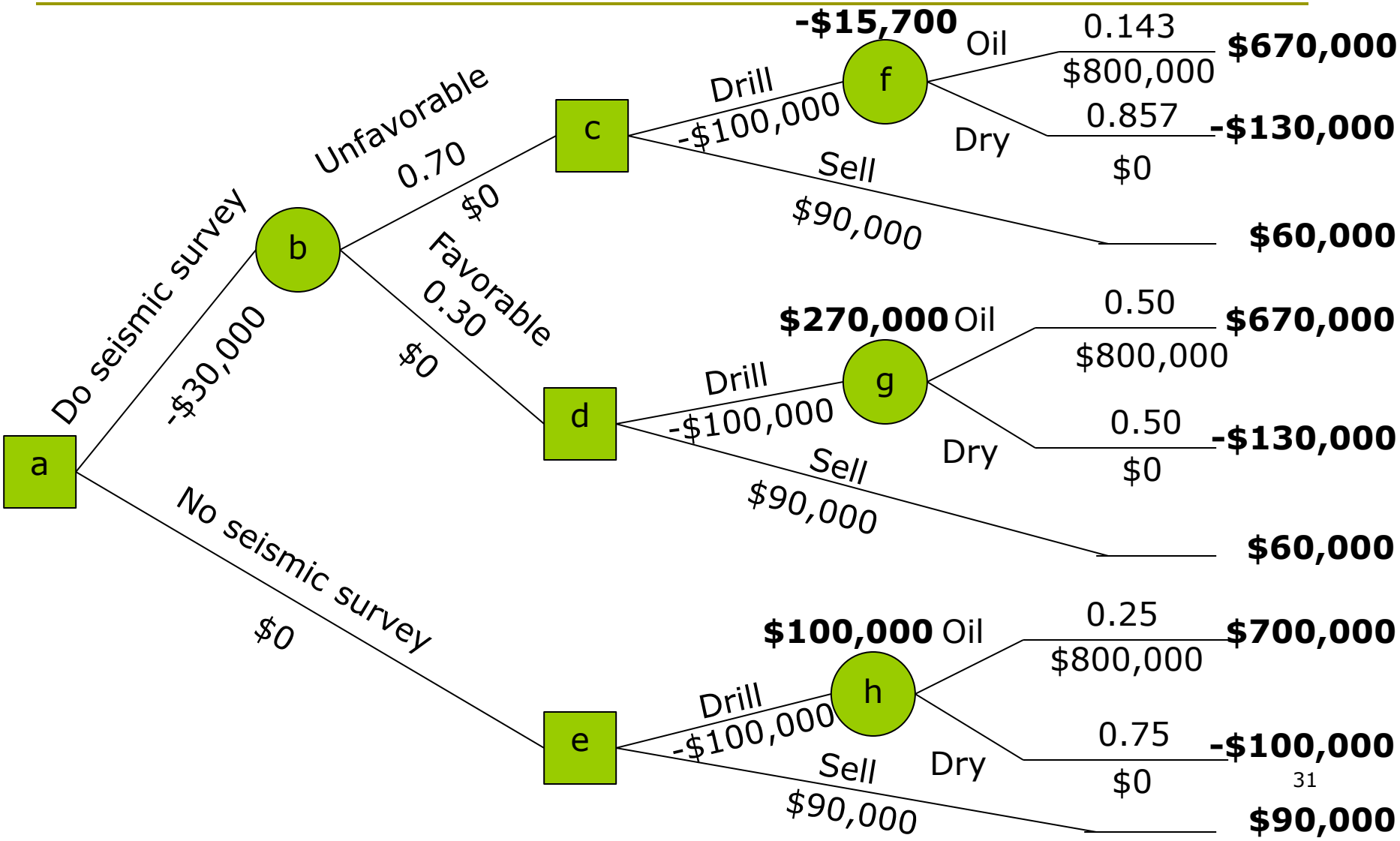
Decision Tree Analysis

- Procedure:
  - 3. For each decision node, compare the expected payoffs of its branches and choose the alternative whose branch has the largest expected payoff. In each case, record the choice on the decision tree by inserting a double dash as a barrier through the each rejected branch.

Decision Tree Analysis

- Consider the rightmost column of nodes.
  - Event nodes *f*, *g*, and *h*, and apply step 2.
  - Expected payoffs:
  - *f*: EP = (1/7)($670,000) + (6/7)(-$130,000) = -$15,700
  - *g*: EP = (1/2)($670,000) + (1/2)(-$130,000) = $270,000
  - *h*: EP = (1/4)($700,000) + (3/4)(-$100,000) = $100,000

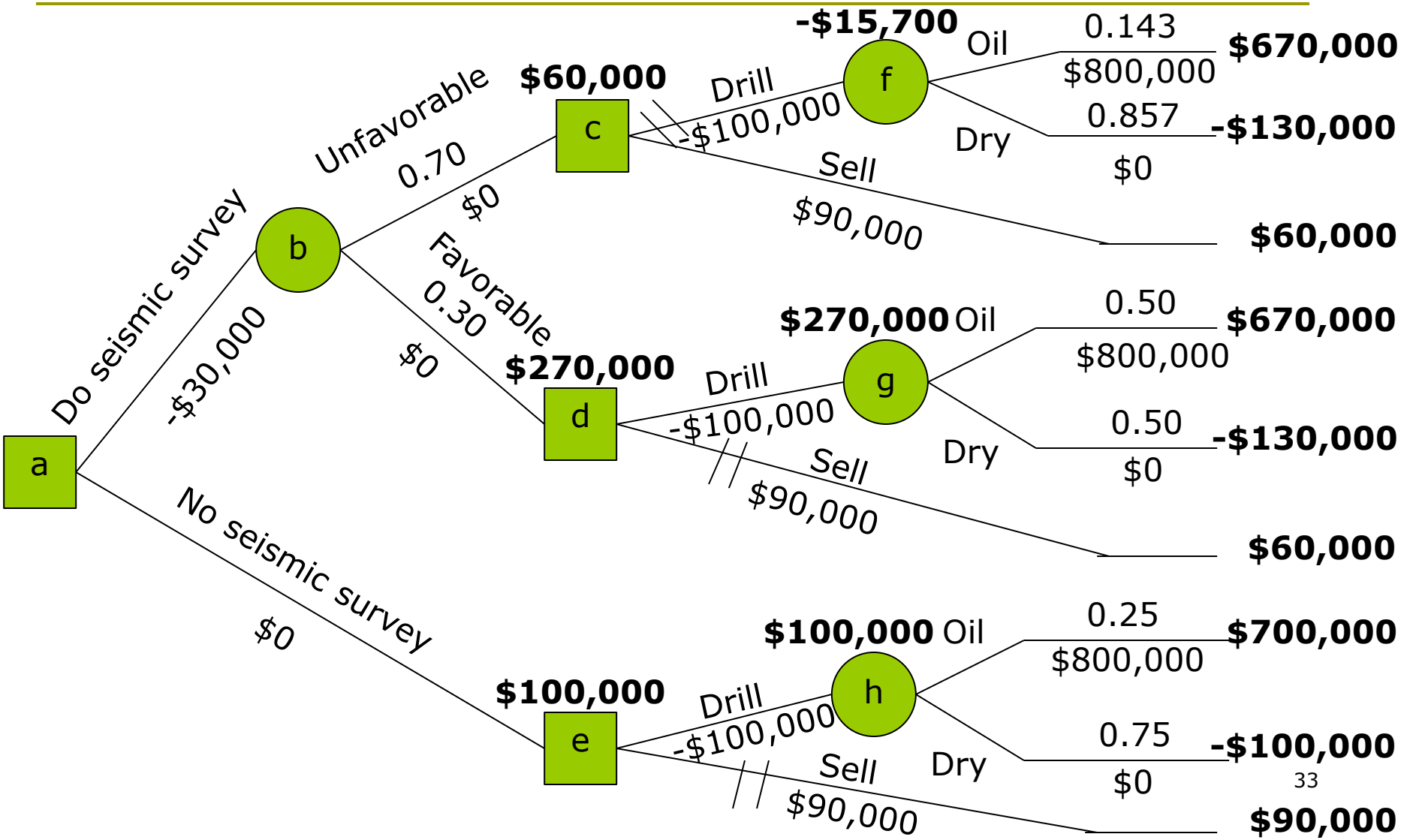
Decision Tree Analysis



Decision Tree Analysis

- Move one column to the left.
  - Decision nodes *c*, *d*, and *e*, and apply step 3.
  - Expected payoffs:
  - *c*: EP(Drill) = -$15,700 < EP(Sell) = $60,000
    - Choose Sell.
  - *d*: EP(Drill) = $270,000 > EP(Sell) = $60,000
    - Choose Drill.
  - *e*: EP(Drill) = $100,000 > EP(Sell) = $90,000
    - Choose Drill.

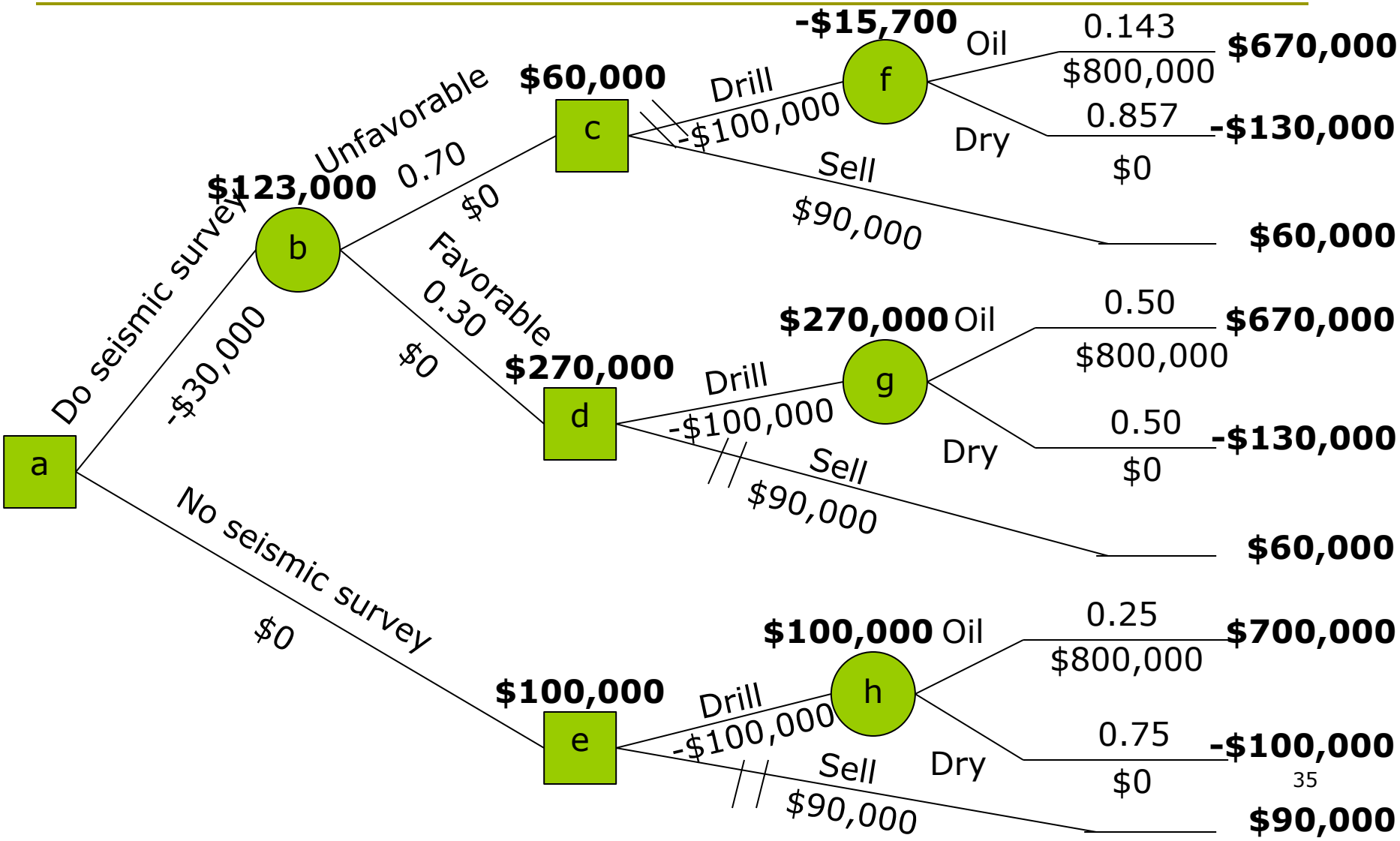
Decision Tree Analysis



Decision Tree Analysis

- Move one more column to the left.
  - Event node *b*, and apply step 2.
  - Expected payoff:
  - *b*: EP = (0.70)($60,000) + (0.30)($270,000) = $123,000

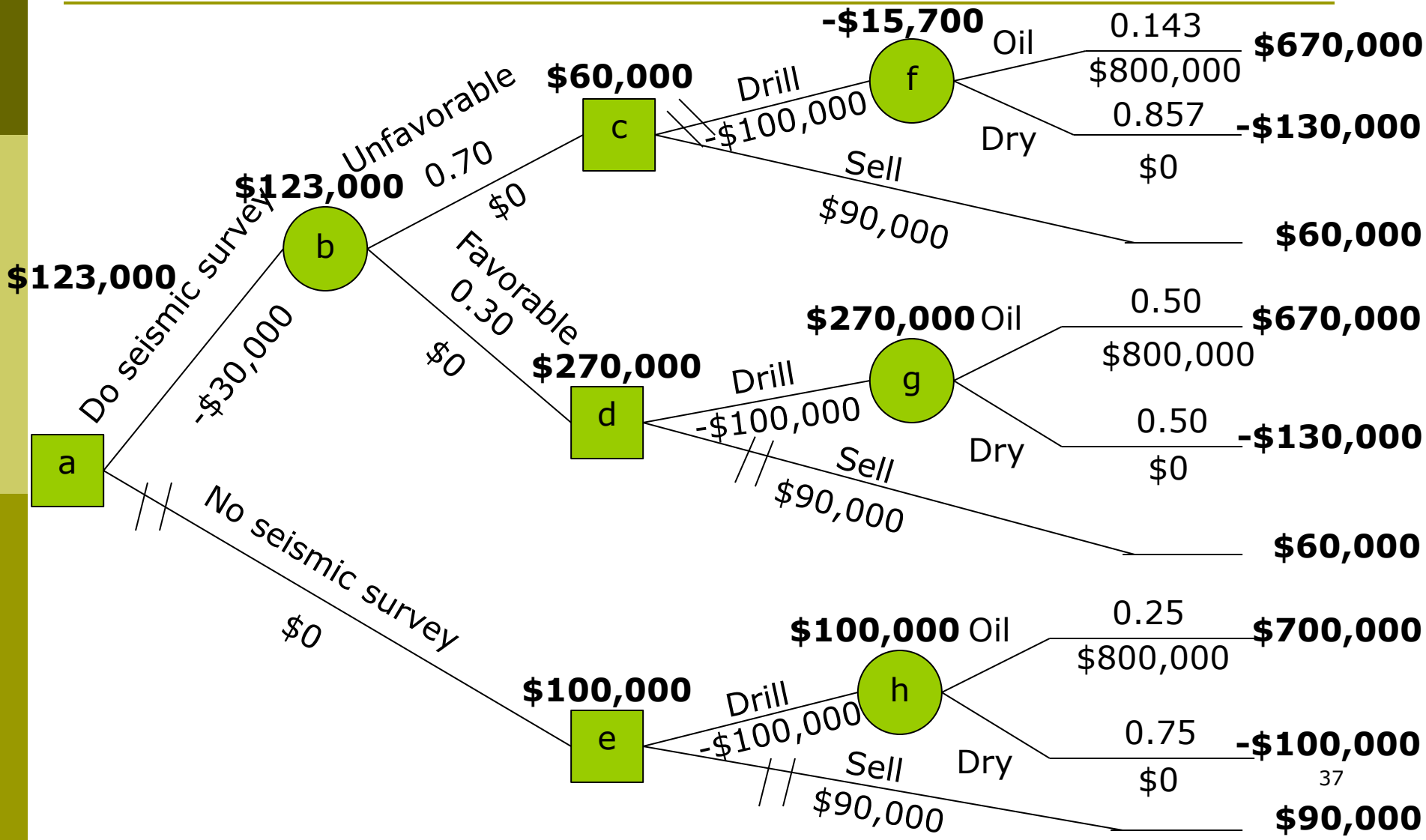
Decision Tree Analysis



Decision Tree Analysis

- Move one more column to the left.
  - Decision node *a*, and apply step 3.
  - Expected payoff:
  - *a*: EP(Do seismic survey) = $123,000 > EP(No seismic survey) = $100,000
  - Choose Do Seismic survey.

Decision Tree Analysis



Decision Tree Analysis

- Follow the open path from left to right to achieve the largest possible payoff.
  - Do the seismic survey.
  - If the result is unfavorable, sell the land.
  - If the result is favorable, drill for oil.
  - The expected payoff (including the cost of seismic survey) is $123,000.

Utility Theory

- Relax the assumption that the expected payoff in monetary terms is the appropriate measure of the consequences of taking an action.

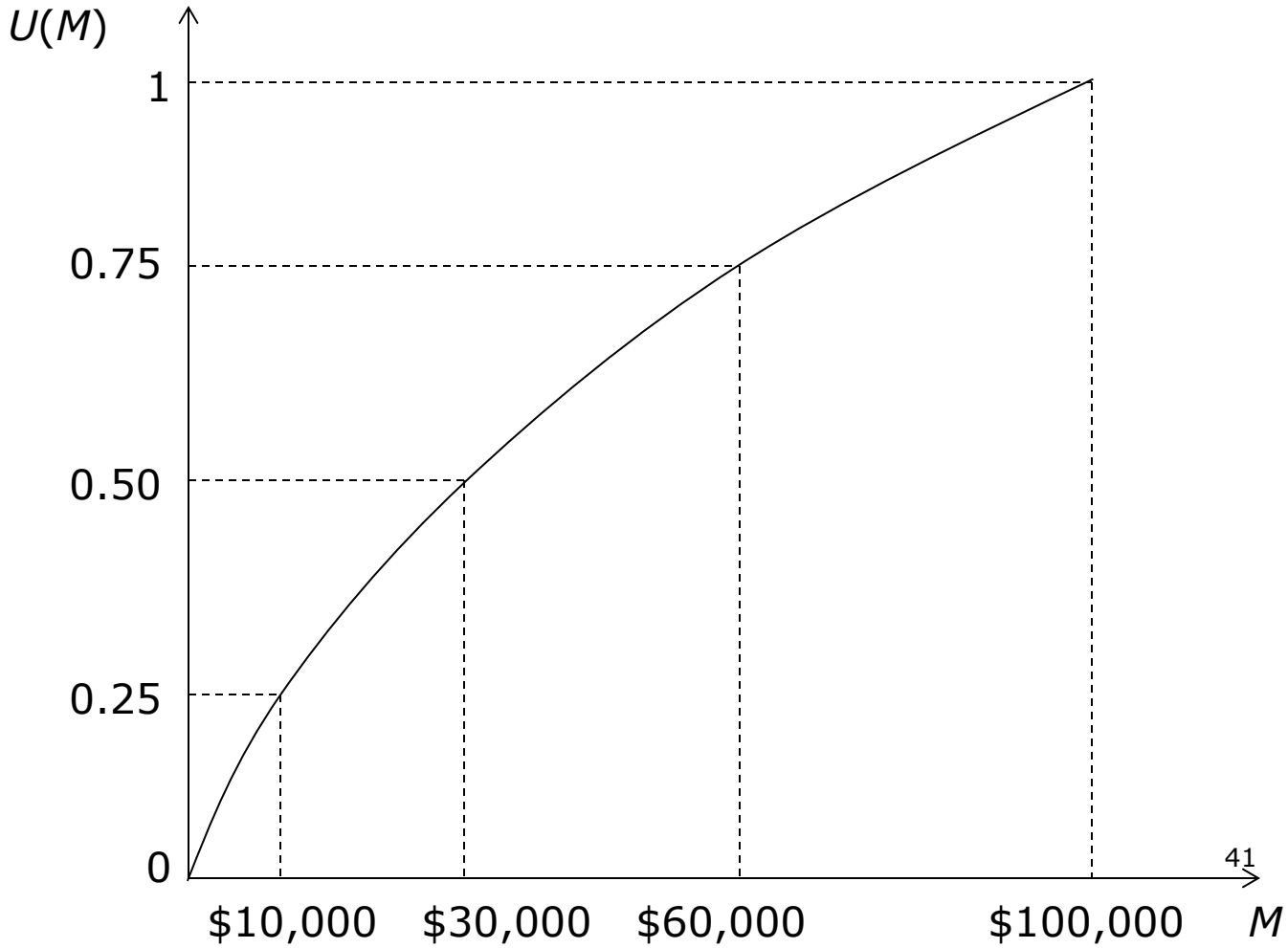
- Examples:
  - An individual is given the choice of (1) accepting a 50:50 chance of winning $100,000 or nothing or (2) receiving $40,000 with certainty. Most people would prefer the $40,000 even though the expected payoff on the 50:50 chance of winning $100,000 is $50,000.
  - People buy insurance even though it is a poor investment from the viewpoint of expected payoff.

Utility Theory

- Do the examples invalidate Bayes' Decision Rule?
  - Answer is no.
- There is a way of transforming monetary values to an appropriate scale that reflects the decision maker's preferences.
  - The scale is called the utility function for the money.

Utility Theory

- Typical utility function $U(M)$ for money M.



Utility Theory

- Typical utility function $U(M)$ for money M.
  - An individual having this utility function would value obtaining $30,000 twice as much as $10,000 and would value obtaining $100,000 twice as much as $30,000.
- Decreasing slope of utility function is referred to as having a decreasing marginal utility for money.
  - Risk-averse individual.
- Increasing slope of utility function is referred to as having an increasing marginal utility for money.
  - Risk-seeker individual.
- Individuals who prize money at its face value are called risk-neutral.

Utility Theory

- Fundamental property:
  - Under the assumptions of utility theory, the decision maker's utility function for money has the property that the decision maker is indifferent between two alternative courses of action if the two alternatives have the same expected utility.

- Set the value of $U(M)$ for two values.
  - Set $U(M) = 0$ for the smallest value of M.
  - Set $U(M) = 1$ for the largest value of M.

Utility Theory

- Equivalent Lottery Method.
  - 1. Determine the largest potential payoff, M = *maximum*, and assign it some utility.

    $U(maximum) = 1.$
  - 2. Determine the smallest potential payoff, M = *minimum*, and assign it some utility smaller than in step 1.

    $U(minimum) = 0.$
  - 3. To determine the utility of another potential payoff M, the decision maker is offered the following two hypothetical alternatives:

    A1: Obtain a payoff of maximum with probability p
    Obtain a payoff of minimum with probability $1 - p$

    A2: Definitely obtain a payoff of M.

Utility Theory

- Equivalent Lottery Method.
  - Question to the decision maker:

    What value of p makes you indifferent between these two alternatives?

    The resulting utility of M then is

    $U(M) = pU(maximum) + (1 - p)U(minimum)$

    $U(M) = p$, if $U(minimum) = 0$, $U(maximum) = 1$

  - When the decision maker's utility function for money is used to measure the relative worth of the various possible monetary outcomes, Bayes' decision rule replaces monetary payoffs by the corresponding utilities.
  - Therefore, the optimal action (or series of actions) is the one which maximizes the expected utility.

Utility Theory Example

- Worst-case scenario would be to do the seismic survey for $30,000 and then still lose $100,000 by drilling when there is no oil.
- Best case: striking oil and earn $700,000.

- Set $U(minimum) = U(-\$130{,}000) = 0$
- Set $U(maximum) = U(\$700{,}000) = 1$

- Apply equivalent lottery method to determine the utility for another of the possible monetary payoffs.

Utility Theory Example

- Questions asked the decision maker:
  - Suppose you have only the following two alternatives.

    A1: Obtain a payoff of \$700,000 with probability p

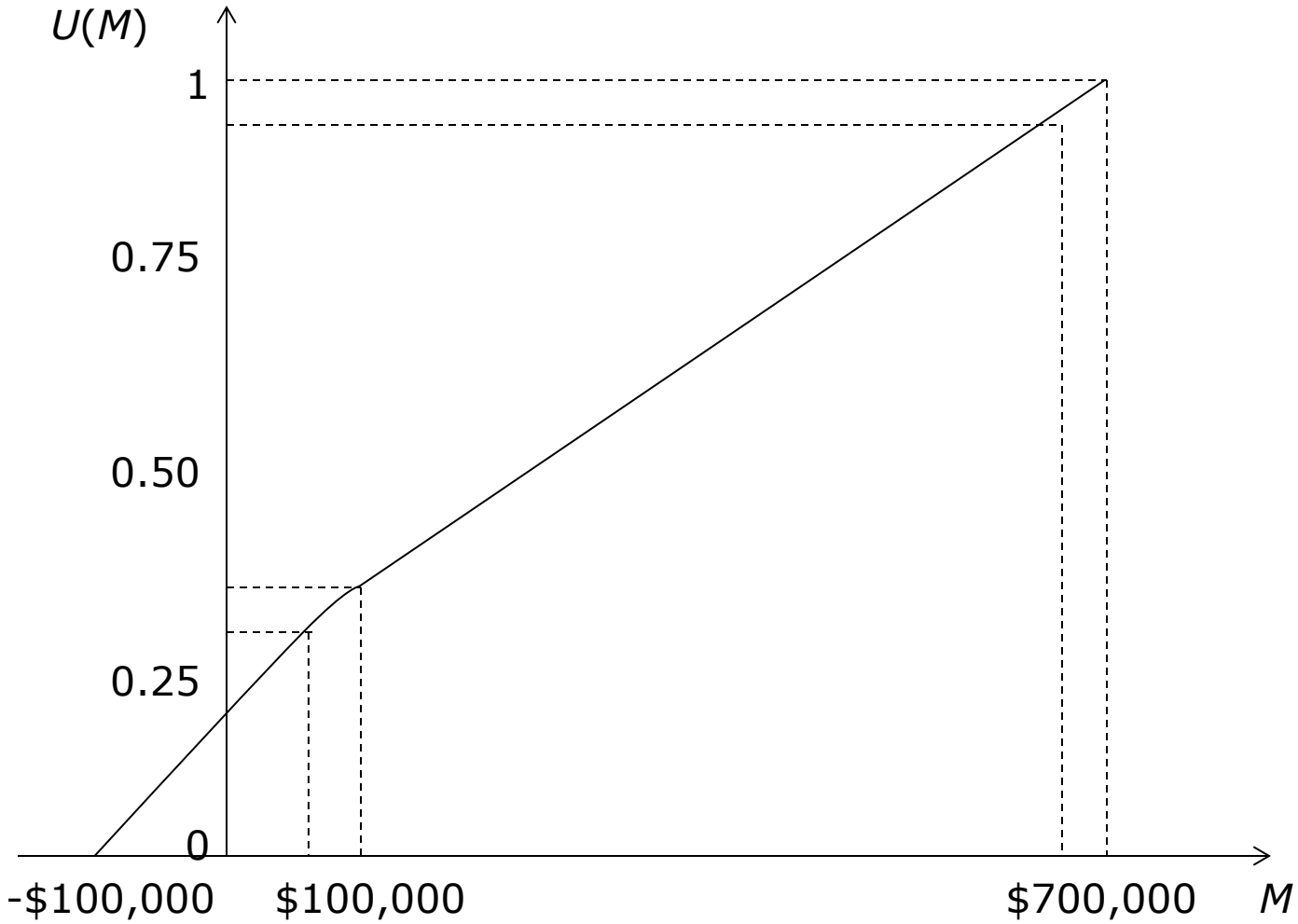
    Obtain a payoff of -\$130,000 with probability $1 - p$

    A2: Definitely obtain a payoff of \$90,000

    What value of p makes you indifferent between these two alternatives?
  - Decision maker's choice: $p = 1/3$, so $U(\$90{,}000) = 0.333$
- Method applied for $M = -\$100{,}000$
  - Decision maker's choice: $p = 1/20$, so $U(-\$100{,}000) = 0.05$
- Draw the utility function and determine
  - $U(\$60{,}000) = 0.30$ and $U(\$670{,}000) = 0.97$

Utility Theory Example

- Utility function $U(M)$ for money M.



Utility Theory Example

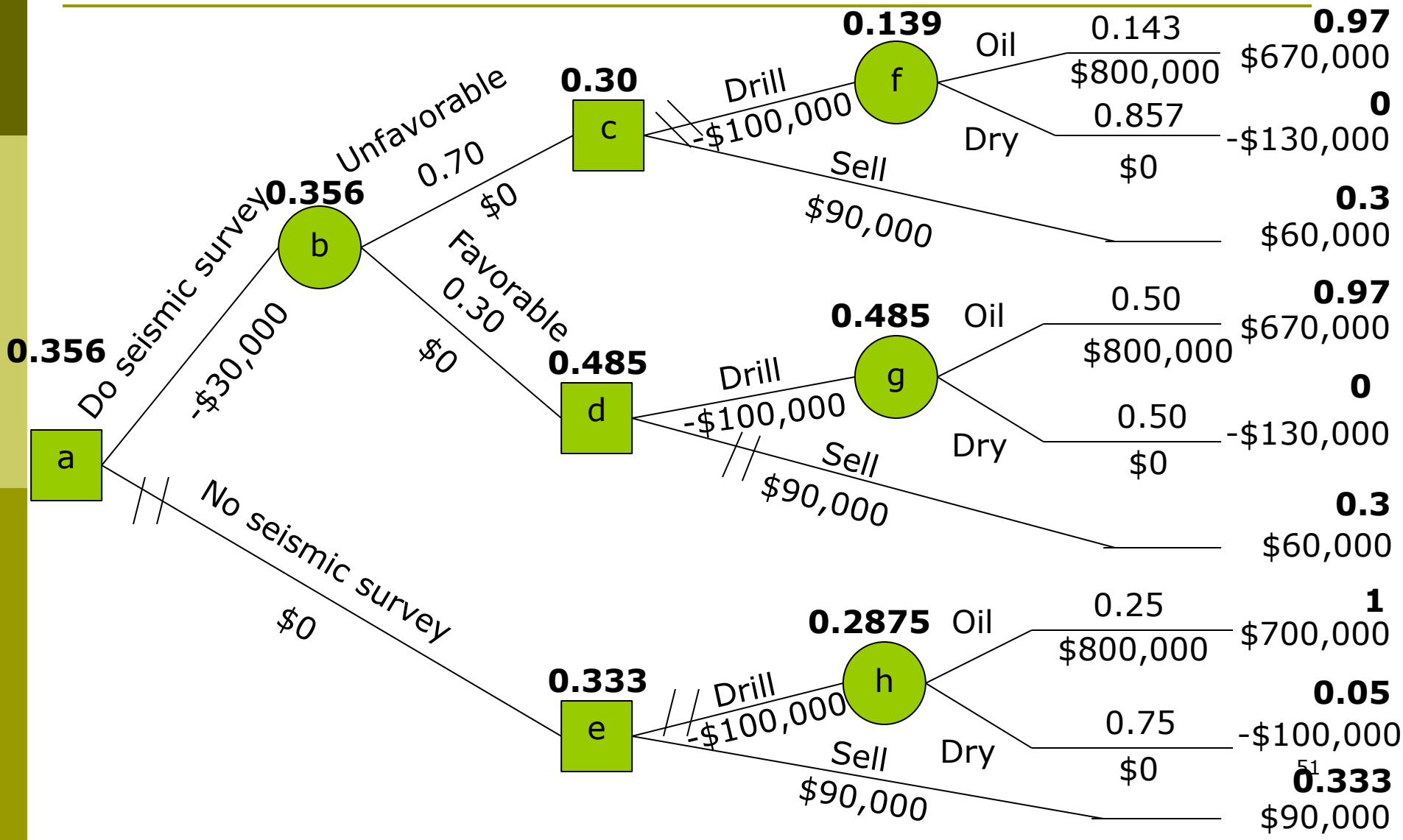
- Utilities for the problem.

Monetary payoff	Utility
-$130,000	0
-$100,000	0.05
$60,000	0.30
$90,000	0.33
$670,000	0.97
$700,000	1

Utility Theory Example

- Procedure for using a decision tree to analyze the problem is identical to that already described except for substituting utilities for monetary payoffs.
  - The value obtained to evaluate each node of the tree is the expected utility.
  - Optimal decision selected by the Bayes' decision rule maximizes the expected utility for the overall problem.

Decision Tree Analysis



Utility Theory Example

- Expected utilities lead the same decisions at nodes *a*, *c*, and *d*.
- Decision at node *e* switches to sell instead of drill.
- Backward procedure still leaves node *e* on a closed path.
- Overall optimal policy remains the same.
  - Do the seismic survey.
  - Sell if result is unfavorable.
  - Drill if result is favorable.

- Moderately risk-averse policy.
  - For a somewhat more risk-averse policy, the optimal solution would switch to the more conservative approach of immediately sell the land (without seismic survey).